

Yanbo Pan

+1-608-960-5479 | war8sk@virginia.edu | <https://yanbopanpi.github.io>

 [LinkedIn](#) |  [yanbopanpi](#) |  [ORCID](#)

530 McCormick Road | Charlottesville, VA, 22904

RESEARCH INTERESTS

Observational Near-Field Cosmology; Dwarf Galaxies; Stellar Dynamics; Stellar Populations; Large Sky Surveys; Computational and Machine Learning Applications in Astrophysics, Astrostatistics

EDUCATION

- **University of Virginia** 2025 – 2030 (*Expected*)
PhD in Astronomy Charlottesville, VA
- **University of Michigan** 2022-2025
Bachelor of Science: Astronomy & Astrophysics (with High Honors), Statistics, Interdisciplinary Physics Ann Arbor, MI
 - GPA: 3.99/4.00 (High Distinction)
- **University of Wisconsin** 2021-2022
Physics & Astronomy Madison, WI
 - GPA: 4.00/4.00

HONORS & AWARDS

1. Provost's Fellowship, University of Virginia (\$32,500) 2025–2030
2. Double Hoo Research Award, University of Virginia (\$6,000) 2026
3. University Graduate Fellowship, Penn State University (Declined) 2025
4. Homer F. Braddock/Nellie H. and Oscar L. Roberts Fellowship, Penn State University (Declined) 2025
5. Jerome and Isabella Karle Physical Sciences Award, University of Michigan (\$2,000) 2025
6. Outstanding Undergraduate Major Award, Department of Statistics, University of Michigan 2025
7. Graduation Speaker, Department of Statistics, University of Michigan 2025
8. Dorrit Hoffleit Undergraduate Research Scholarship, Yale University (\$4,600) 2024
9. ERES IX Conference Travel Award, Cornell University (\$500) 2024
10. James B. Angell Scholar, University of Michigan 2023–2025
11. University Honors, University of Michigan 2022–2025
12. Dean's List, University of Wisconsin 2021–2022

PUBLICATIONS

J=JOURNAL, P=IN PREP, T=THESIS

- [J.1] Fournier-Tondreau, M., Pan, Y., et al. (2025). **Transmission spectroscopy of WASP-52 b with JWST NIRISS: water and helium atmospheric absorption, alongside prominent star-spot crossings.** *MNRAS*.
- [P.1] Pan, Y., et al. (2025). **JWST MEP - A World Under Spotty Starlight: Detection of CO₂ and H₂O in the Hot Saturn WASP-52b with JWST NIRSpec G395H.** (In Review).

SCIENCE TALKS

- APOGEE's Galactic Odyssey**, Teruel, Spain Aug 2026
Sgr dSph Unveiled: Metallicity Distributions and 3D Kinematics in the Main Body of a Disrupting Satellite Galaxy
- Bob Rood Memorial Symposium**, Charlottesville, VA April 2026
HARLEE: Your Next RV Observing Partner with theJoker
- UVa Astronomy Journal Club**, Charlottesville, VA December 2025
JWST MEP - A World Under Spotty Starlight: Atmospheric Forecast for the Hot Jupiter WASP-52b
- AAS 245**, National Harbor, MD January 2025
Exploring the Chemical Recipes of a Hot Jupiter Behind the Shade of Spotty Starlight
- Yale Summer Research Symposium**, New Haven, CT July 2024
Riding Two Dark Horses: A Spectroscopic Study of Ultra-faint Milky Way Satellites
- ERES IX @ Cornell**, Ithaca, NY July 2024
The Effect of Stellar Features on Atmospheric Retrievals of WASP-52b's Transmission Spectrum
- Student Astronomical Society Research Talk**, Ann Arbor, MI September 2023
Modelling Binary Orbits in the WIYN Open Cluster Survey - The alternative evolutionary tracks for binary stars

POSTERS

2nd Annual Cosmic Horizons Conference , Charlottesville, VA	July 2026
Less is More: Constraining Binary Orbits using the Helper for Analyzing RV with Limited Epochs and Error	
AAS 244 , Madison, WI	June 2024
Modelling Binary Orbits in the WIYN Open Cluster Survey	
University of Michigan Undergraduate Research Fair , Ann Arbor, MI	April 2024
Mining Ultra-Faint Galaxies in the Local Group: Are Missing Satellites All Found?	
University of Michigan Undergraduate Research Fair , Ann Arbor, MI	April 2023
Uncovering Ultra-Faint Galaxies in the Local Group	

RESEARCH EXPERIENCES

Graduate Research, Astronomy Department University of Virginia, VA	2025 – Present
<i>Research Assistant (Advisors: Steve Majewski & Nitya Kallivayalil)</i>	
Chemo-dynamical analysis of resolved stellar populations in Sagittarius Dwarf Galaxy using Gaia EDR3	
Undergraduate Research, Astronomy Department University of Michigan, MI	2024 – Present
<i>Research Assistant (Advisor: Ryan MacDonald)</i>	
Multidimensional atmosphere analysis of WASP-52b using JWST Cycle 2 NIRISS & NIRSpec transmission spectra	
Undergraduate Research, Astronomy Department Yale University, CT	2024 Summer
<i>Dorrit Hoffleit Scholar (Advisor: Marla Geha)</i>	
A spectroscopic study of two ultra-faint Milky Way satellites' dynamics using Keck DEIMOS spectroscopy	
Undergraduate Research, Astronomy Department UW-Madison, WI	2023 – 2024
<i>Research Assistant (Advisor: Robert Mathieu)</i>	
Evaluate the potential of theJoker in binary orbit modeling for the WIYN Open Cluster Survey	
Undergraduate Research, Astronomy Department University of Michigan, MI	2023 – 2024
<i>Research Assistant (Advisor: Eric Bell)</i>	
Mining M31 ultra-faint dwarf satellites in the DECam Local Volume Exploration Survey DR2 footprint	

OBSERVING EXPERIENCES

APO 3.5m/DSSI	1 night
---------------	---------

STUDENT MENTORSHIP

Emma Breier	(Structure Finding; UVa)	2026
Nikhil Limgala	(Binary orbits modeling; UVa)	2026
Ava Love	(General Advising; UVa)	2026
James McDowell	(Binary orbits modeling; UW-Madison)	2025

TEACHING EXPERIENCES

Teaching Assistant, ASTR 3130: Observational Astronomy	Spring 2026
<i>Graded lab reports; led Python and data analysis discussion sessions/labs on observational techniques; held weekly office hours</i>	

OUTREACH TALKS

Astronomy on Tap , Charlottesville, VA	January 2026
Hard-to-Find Cosmic Seasoning in Our Galactic Boiling Soup	
Graduation Remarks, Department of Statistics , Ann Arbor, MI	May 2025
From Statistical Toolbox to the Expanding Universe	
Student Astronomical Society Outreach Talk , Ann Arbor, MI	April 2025
Dwarf Galaxies: Hidden Gems in Our Cosmic Neighborhood	
Year of the Dragon Planetarium Talk , Ann Arbor, MI	February 2024
The Great Race - The Origin of the Chinese Zodiac	
University of Michigan Physics 391 Talk , Ann Arbor, MI	October 2023
The Missing Satellite Problem	

SCIENCE COMMUNICATION & PUBLIC OUTREACH

Scientific blogger AstroPie@RedNote

Active member of Scientist Network for Advancing Policy (SNAP)

Doghouse telescope operator @ UVa McCormick Observatory

Graduate reviewer for ASPIRE program @ UVa

Undergraduate mentor of the U-M Student Astronomical Society (SAS) UFD group

Undergraduate co-representative of the U-M Astronomy Climate committee

Guest writer of the Astrobites Undergraduate Research Series

Volunteer tutor @ the Greater University Tutoring Service (GUTS)

Community speaker of the Chinese New Year Show @ the U-M Planetarium and Dome Theater

Travel agent summer intern @ China CYTS Tours Holding Co., Ltd

ASTRONOMY DATA SCIENCE SKILLS

- **Proficient:** Python, R, Markdown, Jupyter, $\LaTeX$
- **Intermediate:** SQL, MESA-web, Cloud Computing, Unix Shell, WSL, Version Control
- **Familiar:** C++, HTML/CSS, Docker, MATLAB, DS9
- **Datasets:** DECam Local Volume Exploration Survey (DELVE), ESA - Gaia, WIYN Open Cluster Survey (WOCS), JWST Transmission Spectra (NIRISS & NIRSpec), Keck DEIMOS Spectra
- **Packages:** pandas, numpy, matplotlib, seaborn, scikit-learn, PyTorch, TensorFlow, scipy, sqlite3, emcee, pymultinest, pymc3, theJoker, POSEIDON, ezpadova, gala, fitsio, astropy, astroquery

PROFESSIONAL EXPERIENCES

- **Member of the Astrobiology Team** 2022 - 2023
MRover | *University of Michigan, MI* 
 - Developed chemical tests for life detection and collaborated on implementation for the University Rover Challenge
- **Museum Docent** 2021 - 2022
The L.R. Ingersoll Physics Museum | *UW-Madison, WI* 
 - Facilitated physics demonstrations and conducted educational events during the Wisconsin Science Festival
- **Part-time Tutor** 2021 - 2022
The Jingrui Academy of Science | *Ningbo, China*
 - Provided academic tutoring in math, physics, and economics to IB students through customized instruction